

Dr Waldon Hendricks's notes 2026-07-27 17:57(GMT+2:00)

Topic

- Core focus: explaining the **design phase** of the Capstone project — what it requires, why it matters, how it is structured, and how it will be marked
 - Secondary focus: clarifying the **live document lifecycle** across all four submission phases and providing **stream-specific design examples** to guide students and supervisors
-

What was discussed

AI Tool Usage & Data Privacy

- **Speaker 1** opened with a caution about using AI note-taking tools in class sessions, noting that data entered into external AI tools is sent to outside servers ¹
 - Emphasized that while no current policy blocks specific AI tools, future policies may restrict their use in sensitive or confidential discussions ¹
 - Clarified that personal notes taken by the lecturer would not be sent to a central server, distinguishing them from third-party AI tools ²
-

Purpose of the Session

- **Speaker 1** framed the session as one of the most important of the semester, focused specifically on guiding students through their **first submission: the design phase** ³
 - Supervisors were welcomed; those absent were directed to the recording
 - Rubric details were to be shared with supervisors so they can assess whether students are on track without being overburdened ³
-

Why the Design Phase Matters

- **Speaker 1** used the analogy of an **architect's blueprint**: you cannot build what you have not designed 4
 - The design document serves as an **anchor** connecting all subsequent phases — testing, demonstration, and final report 4
 - A weak design produces a "fragile capstone" — insufficient blueprints mean the build cannot be properly evaluated or traced 5
 - The design phase forces students to **think before they code, configure, or train** a model 4
 - Marks (100 total) are allocated across: design documents, diagrams, prototypes/mockups, and requirements traceability 5
-

Live Document Lifecycle (Four Phases)

- **Speaker 1** outlined the full document growth cycle:
 - **Phase 1 – Design Phase**: High-level and detailed design documents, diagrams, prototypes, traceability matrix
 - **Phase 2 – Testing Phase**: Test plans, test results, revised design sections, updated traceability 6
 - **Phase 3 – Demonstration Phase**: 10-minute live demo + 5-minute Q&A; panel includes lecturer, external moderator, and stakeholders if applicable; shows design-vs-built comparison 6
 - **Phase 4 – Final Report**: Fully integrated report, reflection on design decisions, lessons learned 7
 - The document is **continuously updated** — each phase builds on the previous submission rather than starting fresh 7
 - Original timeline started **19 June**; current session date noted as **2 August**, reflecting the adjusted schedule 8
-

Defining a Real-World Problem (Evidence Requirements)

- **Speaker 1** identified the **most common capstone failure**: a project without a grounded, evidence-based problem ⁷
 - A valid problem must meet the following criteria:
 - **Real-world impact**: Affects actual people, organizations, or systems — not a "toy problem" ⁹
 - **Quantified**: Supported by statistics, reports, or studies — not personal opinion ⁹
 - **Literature-backed**: Minimum **3–5 peer-reviewed sources** (IEEE, ACM, Scopus, Google Scholar) ^{9 10}
 - **Specific target group**: E.g., "small retailers in South Africa" or "SMEs" rather than "all users" ¹¹
 - **Gap justification**: Must show what exists today vs. what the project provides ¹¹
 - **Examples of invalid problem statements** explicitly flagged: ^{10 11}
 - "I want to build an app" — not a problem
 - "Technology is advancing fast" — too vague
 - "A friend complained about this" — anecdotal, not published research
 - "It's a cool system" — no evidential purpose
 - **Acceptable evidence sources**: Peer-reviewed journals, government/industry reports, ISO standards, institutional legislation — all cited in **Harvard style** ¹²
 - Survey-based evidence was noted as possible but discouraged due to **ethics complications** ¹⁰
-

Design Phase Template Structure

- **Speaker 1** walked through the required sections of the design document template (available on Blackboard): ¹²
 - Introduction
 - Problem scope
 - Objectives (the "evidence pack")

- Design overview
 - Approach and justification
 - Design documents (diagrams, visual aids, prototypes, mockups)
 - Requirements traceability matrix
 - Stream-specific section
 - Design evaluation
 - Conclusion
 - References (minimum 5 sources)
 - Appendices (if applicable)
 - Self-assessment checklist
- The **problem statement section** will be reviewed by the supervisor (domain expertise), not marked directly in the rubric ¹²
 - A **self-assessment Excel checklist** will be uploaded to Blackboard so students can verify alignment with the rubric before submitting ¹³
-

Stream-Specific Design Examples

Application Development — Township SME Inventory System

- **Evidence:** Over 70% of South African informal SMEs operate without a digital inventory system, leading to estimated annual stock losses of R2.1 billion; manual stock management causes overstocking, spoilage, and cash flow constraints ¹⁴
- **Tech stack:** React Native (mobile), Node.js, Azure REST API, PostgreSQL ¹⁵
- **Diagrams:** ERD with 6 entities (product, supplier, transaction, category, user, alerts); UML sequence diagram for stock update flow; UI wireframes (4 primary screens) ¹⁵
- **Prototype:** Figma clickable mockup of 4 screens ¹⁵
- **Traceability example:** Requirement — offline mode; Justification — rural connectivity constraints (GSMA data) ¹⁶

Communication Networks — Secure Campus Wi-Fi Redesign

- **Evidence:** Campus network downtime at South African universities averages 4.2 hours/week (Tenet annual report); unencrypted networks expose 12,000+ devices to lateral movement attacks; impacts 40,000+ students ¹⁶
- **Tech stack:** Core redundant Cisco 9500, distribution VLAN segmentation, access layer protocols ¹⁷
- **Diagrams:** Network topology via Packet Tracer, Visio, or diagrams.net ¹⁷
- **Simulation tools:** Cisco Packet Tracer or NS3 ¹⁷
- **Traceability example:** Requirement — 99.5% uptime SLA; Justification — redundant core switches with HSRP ¹⁸

AI & Data Science — Early Crop Disease Detection

- **Evidence:** Crop diseases cause 20–40% of global food production losses annually; Sub-Saharan African smallholder farmers lose ~25% of yield to late blight and leaf rust; early visual detection via smartphone can reduce losses by up to 30% ¹⁸
- **Model:** ResNet architecture; PlantVillage dataset (~54,000 images, 38 classes); fine-tuned model ¹⁹
- **Data pipeline:** Raw images → augmentation (flip, brightness) → split → training ¹⁹
- **Evaluation:** F1 score, confusion matrix ¹⁹
- **Prototype:** Jupyter Notebook running accuracy/validation; matplotlib for mockups ¹⁹
- **Traceability example:** Accuracy requirement mapped to ResNet-50 architecture demonstrating ~99.35% on the same dataset ²⁰

Web Technology — Community Health Clinic Booking Portal

- **Evidence:** 87% of South African public health clinic patients still book via walk-in queues, with average wait times of 3.4 hours (Dept. of Health report); WhatsApp-based appointment reminders reduced missed appointments by 40% in a pilot study ²⁰
- **Tech stack:** Progressive Web App (PWA), Express.js, REST API, MongoDB ²¹
- **Diagrams:** Figma wireframes (5 pages, 2 personas defined) ²¹

- **Traceability example:** SMS integration via Twilio (trial subscription noted; warned against entering credit card details) ²¹

Information Systems — School Learner Records Management System

- **Evidence:** 60% of South African public schools manage learner records via paper-based systems, with an average data retrieval time of 22 minutes per record; manual record keeping results in 18% administrative error rates impacting Grade 12 certification compliance ²²
 - **Tech stack:** React (frontend), Spring Boot REST backend, MySQL ²³
 - **Database schema:** 8 tables — learner, guardian, enrollment, grade, subject, teacher, report, and audit/lock ²³
 - **Traceability example:** Compliance requirement — audit lock table + data masking module to prevent data breaches on student records ²³
-

Common Mistakes That Cost Marks

- **Speaker 1** explicitly listed the following errors to avoid: ²⁴ ²⁵
 - No citations in the problem statement
 - No diagrams, or diagrams without numbered captions
 - High-level design consisting of only one paragraph (must include diagrams, component descriptions, and a tech stack table)
 - Traceability matrix left vague or blank
 - Prototype with only one screenshot and no annotation explaining what each screen validates
 - Reusing the same diagram multiple times in the document
 - Stream-specific section left incomplete
 - Problem scope too small or too broad — must target **one specific real-world problem** with one target group ²⁵
 - Treating the design phase as a one-off submission rather than a living document ²⁶

Rubric Breakdown (100 Marks)

- **Speaker 1** walked through the performance bands: **Excellent** → **Proficient** → **Developing** → **Not Yet** (below 50% on a given criterion) 27
- Mark allocation across five criteria: 28 29
 - a. **High-Level Design Document** — fully described architecture, components, and tech stack
 - b. **Detailed Design Document** — explanation of how the design is constructed
 - c. **Diagrams & Visual Aids** — architecture/topology/system diagrams, sequence/flow diagrams, ERD/wireframes
 - d. **Prototypes & Mockups** — screenshots, simulation outputs, annotated functionality and design validation
 - e. **Requirements Traceability** — each row must include an ID, design element, and justification
- **Evaluation & Academic Quality** (holistic criterion): technical depth across all three dimensions, academic writing quality, referencing, and overall presentation 29
- **Distinction-level work** requires strong performance across: problem statement, design documents, diagrams, prototypes, traceability, evaluation, and conclusion 26

Supervisor Guidance Notes

- Supervisors are expected to assess whether students **understand their domain** and are on track — not to push too hard on research methodology 30
 - The capstone operates within **design science research** principles, not a full research methodology framework 30
 - Five streams in the capstone: **Application Development, Communication Networks, AI & Data Science, Web Technology, Information Systems** — each with two elective subjects 31
 - Students who wish to design outside their primary stream must discuss the approach with their supervisor 8
-

Takeaway / Understanding

The session established that the **design phase is the foundational blueprint of the entire capstone**, and its quality directly determines the strength of all subsequent phases. Students must ground their projects in **empirically evidenced real-world problems**, produce technically detailed design documents with properly annotated diagrams and prototypes, and maintain the document as a **living artifact** updated throughout the semester.

Action items

- ☐ Download the design phase template from Blackboard and complete all required sections (All students)
- ☐ Define a real-world problem statement supported by **3–5 peer-reviewed sources** with quantified evidence (All students)
- ☐ Produce stream-specific design documents including diagrams (ERD, UML, topology, wireframes, etc.) with numbered captions (All students)
- ☐ Build at least one annotated prototype screenshot or simulation output validating design functionality (All students)
- ☐ Complete the requirements traceability matrix with IDs, design elements, and justifications for each requirement (All students)
- ☐ Complete the self-assessment checklist Excel sheet before submission to verify rubric alignment (All students)
- ☐ Upload presentation slides, notes, and rubric Excel sheet to Blackboard (Speaker 1 / Lecturer)
- ☐ Share rubric details and stream-specific guidance with all supervisors (Speaker 1 / Lecturer)
- ☐ Student who requested a 1-on-1 meeting: send MS Teams meeting link to lecturer's email for a 7:30 PM session (Named student)